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Effect of Sea Rabbit (Dolabella auricularia) Extract Syrup Combined with Papaya on Hemoglobin and Serum Ferritin Levels in Stunted Toddlers: A Randomized Pretest–Posttest Controlled Trial

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Summary Stunting remains a major public health problem associated with chronic undernutrition and increased risk of micronutrient deficiencies, particularly iron deficiency anemia. This study aimed to evaluate the effect of sea rabbit (Dolabella auricularia) extract syrup combined with papaya extract on hemoglobin and serum ferritin levels in stunted toddlers. A randomized pretest–posttest controlled trial was conducted involving 50 participants allocated into five groups. Three treatment groups received different proportions of sea rabbit extract and papaya extract, while control groups received Ferlyn syrup or placebo. The intervention was administered every two days for 6 weeks. Hemoglobin and serum ferritin levels were measured before and after the intervention. Hemoglobin levels increased significantly in all treatment groups ($p<0.05$), while no significant changes were observed in the control groups. Serum ferritin levels showed numerical increases but did not reach statistical significance ($p>0.05$). These findings suggest that the combination of sea rabbit extract and papaya may improve hemoglobin levels, although longer intervention duration may be required to affect iron storage.

Key Words hemoglobin, ferritin, papaya, sea rabbit extract, stunting, toddlers

Stunting in children under five years of age remains a major public health concern, particularly in low- and middle-income countries. It reflects chronic undernutrition and is associated with impaired physical growth, delayed motor development, reduced cognitive performance, and increased risk of illness.[1,2] These effects may persist into adulthood, contributing to reduced productivity and long-term health problems.

In Indonesia, the prevalence of stunting remains relatively high despite gradual improvements in recent years. Stunting is not only a manifestation of linear growth failure but is also closely related to micronutrient deficiencies, especially iron deficiency. Children with stunting are more likely to develop anemia, which may further impair growth and development. Hemoglobin serves as an important indicator of oxygen-carrying capacity, whereas serum ferritin reflects body iron stores and is widely used to assess iron status.[3–6]

Iron metabolism is influenced by various factors, including dietary intake, infection, inflammation, and other physiological conditions.[7–9] Previous studies have shown that nutritional interventions and supplementation may improve hemoglobin and ferritin levels in populations at risk of anemia.[9,10] However, the response of iron storage markers such as ferritin may require longer intervention periods compared to hemoglobin.

Sea rabbit (*Dolabella auricularia*) has been reported to contain various bioactive compounds, including flavonoids, alkaloids, terpenoids, saponins, and tannins, which may support metabolic and physiological processes.[11] Papaya is also known to contain nutrients and bioactive components that may contribute to hematopoiesis and improved nutritional status.[12–14] The use of locally available food-based resources may therefore provide a practical and accessible approach to improve nutritional and hematological status in children.

Despite this potential, evidence regarding the effect of sea rabbit extract combined with papaya extract on hemoglobin and serum ferritin levels in stunted toddlers remains limited. Therefore, this study aimed to determine the effect of sea rabbit extract syrup combined with papaya extract on hemoglobin and serum ferritin levels in stunted toddlers.

MATERIALS AND METHODS

This study employed a randomized pretest–posttest controlled trial design to evaluate the effect of sea rabbit extract syrup combined with papaya extract on hemoglobin and serum ferritin levels in stunted toddlers. The study was conducted in a primary health care setting in Indonesia from September 2023 to July 2024.

The study population consisted of stunted toddlers identified from five integrated community health posts within the study area. A total of 50 participants were included and divided into five groups, each consisting of 10 children: three treatment groups, one positive control group, and one negative control group. Participants in the control groups were selected using matching criteria based on age and sex.

Inclusion criteria were stunted and malnourished toddlers, hemoglobin level <11.0 g/dL, absence of acute illness such as respiratory infection or diarrhea, and willingness of parents or guardians to provide informed consent. Toddlers with chronic illness, severe medical conditions, or incomplete participation during the intervention period were excluded.

Participants were allocated into five groups as follows: treatment group 1 received sea rabbit extract 80% combined with papaya extract 20%; treatment group 2 received sea rabbit extract 70% combined with papaya extract 30%; treatment group 3 received sea rabbit extract 60% combined with papaya extract 40%; the positive control group received Ferlyn syrup; and the negative control group received placebo. All formulations were prepared using a honey-based syrup. The intervention was administered every two days for 6 weeks.

Sea rabbit samples were collected from coastal areas in Indonesia and processed into extract after repeated washing to reduce salt content and drying at temperatures below 50˚C without direct sunlight exposure. Papaya extract was prepared from ripe papaya obtained from a local market, which was cleaned, cut, and dried using an electric oven at 40–50˚C before extraction. The preparation of extracts was conducted in a university laboratory.

The allocation of intervention formulations was managed by laboratory personnel. The researchers were not informed of the specific composition assigned to each group until the completion of the intervention, indicating a double-blind procedure.

Data collected included baseline characteristics, dietary intake, history of infectious diseases, and laboratory measurements of hemoglobin and serum ferritin. Dietary intake of energy, protein, and iron was assessed using a semi-quantitative food frequency questionnaire. Hemoglobin and serum ferritin levels were measured before and after the intervention.

Data collection instruments included participant screening forms, informed consent forms, a semi-quantitative FFQ, a digital dietary scale, and laboratory equipment for hematological analysis. Field workers received training prior to data collection, and five trained personnel were involved in the study procedures.

Dietary intake data were analyzed using NutriSurvey, the Indonesian Food Composition Table, and the 2019 Recommended Dietary Allowance. Statistical analyses were performed using SPSS version 20. Univariate analysis was used to describe baseline characteristics. Paired sample *t*-test was used to compare pre- and post-intervention hemoglobin and serum ferritin levels within groups, while independent *t*-test was used to compare differences between groups. Although normality testing indicated some deviation from normal distribution, parametric tests were applied considering the sample size and approximate distribution of the data. A p value <0.05 was considered statistically significant.

This study received ethical approval from a recognized health research ethics committee (Ethical Clearance No. 142/KEPK/IAKMI/XI/2020). Written informed consent was obtained from parents or guardians prior to participation.

RESULTS

A total of 96 toddlers underwent initial screening for hemoglobin levels. Among them, 38 children (39.5%) were found to have anemia (hemoglobin <11.0 g/dL), while 58 children (60.4%) had normal hemoglobin levels.

Of the 50 participants included in the study, 38 children (76.0%) were classified as anemic at baseline.

A significant increase in hemoglobin levels was observed in all treatment groups after intervention.

Although serum ferritin levels increased in all groups after intervention, the changes were not statistically significant.

DISCUSSION

This study demonstrated that administration of sea rabbit extract syrup combined with papaya extract was associated with a significant increase in hemoglobin levels in stunted toddlers. The improvement was consistently observed across all treatment groups, whereas no statistically significant change was identified in the control groups. These findings suggest that the intervention may contribute to improving hematological status in children with nutritional deficiencies.

Hemoglobin concentration is influenced by the availability of essential nutrients required for erythropoiesis, particularly iron, protein, folate, and vitamin B12. The observed increase in hemoglobin may reflect improved nutrient availability or utilization following the intervention. Sea rabbit (Dolabella auricularia) contains bioactive compounds such as flavonoids, alkaloids, terpenoids, saponins, and tannins, while papaya provides nutrients and bioactive components that may support hematopoiesis. The combination of these components may enhance metabolic processes related to erythrocyte production.

In contrast, serum ferritin levels did not show statistically significant changes, although numerical increases were observed. This may be explained by physiological differences between hemoglobin and ferritin, as ferritin reflects iron storage and typically requires a longer duration of intervention. In addition, ferritin levels may be influenced by infection and inflammation, which were not assessed in this study.

The relatively short duration of intervention and small sample size may have limited the ability to detect significant changes in ferritin levels. Despite these limitations, the findings highlight the potential of locally available food-based interventions to improve hemoglobin levels in stunted children. Further studies with longer intervention periods and larger sample sizes are recommended to better evaluate their effect on iron status.

Conflict of Interes

The authors declare that they have no conflicts of interest in relation to this study.

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Table 1. Distribution of Hemoglobin Levels at Screening

Health Post	Anemia n (%)	Normal n (%)	Total n (%)
Melati	12 (52.1)	11 (47.8)	23 (23.9)
Mekar Sari	8 (42.1)	11 (57.8)	19 (19.7)
Lalodati	9 (37.5)	15 (62.5)	24 (25.0)
Abeli D	5 (35.7)	9 (64.2)	14 (14.5)
Sinar Pagi	4 (25.0)	12 (75.0)	16 (16.6)
Total	38 (39.5)	58 (60.4)	96 (100)

Table 2. Baseline Nutritional Intake of Participants

Group	Energy (kcal)	Protein (g)	Iron (mg)	p-value
Treatment 1	1003	37.4	9.2	
Treatment 2	1033	39.0	10.6	
Treatment 3	1008	37.0	10.3	
Control (+)	1040	33.2	8.9	
Control (−)	1119	41.3	8.6	0.254

Table 3. Changes in Hemoglobin Levels Before and After Intervention

Group	Before (Mean ± SD)	After (Mean ± SD)	t-value	p-value
Treatment 1	10.9 ± 0.61	11.7 ± 0.51	-4.43	0.002
Treatment 2	11.1 ± 0.86	11.8 ± 0.57	-3.55	0.008
Treatment 3	10.8 ± 1.23	11.6 ± 0.45	-3.10	0.015
Control (+)	10.9 ± 1.02	10.9 ± 0.90	-0.73	0.486
Control (−)	10.7 ± 0.68	11.2 ± 0.38	-2.31	0.054

Table 4. Comparison of Hemoglobin Levels Between Groups After Intervention

Group	Mean Hb (g/dL)	t-test	p-value
Treatment 1	11.7 ± 0.51	-1.448	0.181
Treatment 2	11.8 ± 0.57	-0.795	0.447
Treatment 3	11.6 ± 0.45	-1.880	0.093
Control (+)	10.9 ± 0.90	-1.621	0.139
Control (−)	11.2 ± 0.38	-1.583	0.148

Table 5. Changes in Serum Ferritin Levels Before and After Intervention

Group	Before (Mean ± SD)	After (Mean ± SD)	t-value	p-value
Treatment 1	27.54 ± 16.82	43.45 ± 28.47	-1.95	0.082
Treatment 2	13.83 ± 7.17	16.75 ± 7.75	-1.78	0.109
Treatment 3	14.22 ± 15.72	19.47 ± 13.54	-1.47	0.174
Control (+)	6.83 ± 4.73	8.72 ± 5.86	-0.92	0.380
Control (−)	9.90 ± 8.51	13.89 ± 9.72	-1.34	0.213

Table 6. Comparison of Serum Ferritin Levels Between Groups After Intervention

Group	Mean Ferritin	t-test	p-value
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Group	Before (Mean ± SD)	After (Mean ± SD)	t-value	p-value
Treatment 1	10.9 ± 0.61	11.7 ± 0.51	-4.43	0.002
Treatment 2	11.1 ± 0.86	11.8 ± 0.57	-3.55	0.008
Treatment 3	10.8 ± 1.23	11.6 ± 0.45	-3.10	0.015
Control (+)	10.9 ± 1.02	10.9 ± 0.90	-0.73	0.486
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